

CYPRESS RIVER, MB, Canada

WMO: 712390

Lat: 49.5622N

Lon: 99.0739W

Elev: 1227

StdP: 14.06

Time Zone: -6.00 (NAC)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.9	-19.2	-30.3	1.0	-23.7	-25.6	1.4	-18.9	28.3	11.3	25.2	16.9	6.5	220	0.669

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.5	86.8	68.2	83.4	67.0	80.4	65.7	72.2	81.9	70.0	79.7	67.9	76.9	10.9	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.9	111.0	78.1	66.5	102.3	76.0	64.5	95.0	73.1	36.8	81.6	34.9	79.8	33.0	76.7	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.0	21.1	18.9	-30.0	94.4	5.2	3.1	-33.8	96.7	-36.9	98.5	-39.8	100.3	-43.6	102.6
			-30.2	76.9	5.2	2.1	-34.0	78.4	-37.0	79.7	-39.9	80.9	-43.7	82.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.0	6.6	6.8	22.5	38.2	52.7	62.4	67.0	65.2	55.9	41.7	25.0	9.6
	DBStd	24.73	15.15	13.56	14.59	10.98	9.17	5.75	5.16	6.44	8.57	9.43	12.69	14.07
	HDD50	6198	1346	1211	854	372	78	1	0	0	38	294	751	1253
	HDD65	10117	1811	1631	1316	804	395	113	38	80	290	722	1199	1718
	CDD50	1808	0	0	3	18	161	373	526	472	216	38	1	0
	CDD65	252	0	0	0	0	12	35	99	86	19	1	0	0
	CDH74	2850	0	0	0	2	254	375	958	990	255	16	0	0
	CDH80	755	0	0	0	0	65	64	219	332	72	3	0	0
Wind	WSAvg	9.0	9.5	8.9	9.2	10.3	10.1	8.7	7.5	7.4	8.7	9.7	9.6	8.8
Precipitation	PrecAvg	20.1	0.7	0.7	0.9	1.2	3.0	3.5	2.9	2.3	1.6	1.5	0.8	1.0
	PrecMax	26.7	1.7	2.4	2.1	2.4	6.5	7.4	6.1	4.2	3.9	4.7	3.1	1.7
	PrecMin	15.6	0.0	0.0	0.0	0.2	0.9	1.5	1.0	0.6	0.2	0.1	0.1	0.2
	PrecStd	3.1	0.4	0.6	0.6	0.6	1.2	1.3	1.1	1.0	0.8	1.0	0.7	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.8	39.6	64.3	73.0	86.9	85.3	88.8	92.7	87.7	75.4	59.4	40.4
		MCWB	34.6	35.1	51.8	52.3	60.9	67.4	72.9	69.4	66.6	59.0	49.8	35.1
	2%	DB	36.1	35.8	54.9	66.2	80.3	81.4	84.9	87.6	81.0	68.3	51.2	34.9
		MCWB	32.9	33.1	46.0	49.6	58.1	64.4	70.6	67.9	63.0	53.8	43.7	32.0
	5%	DB	32.8	32.3	46.8	61.6	75.9	78.2	82.3	83.3	75.8	62.6	45.7	31.6
		MCWB	30.9	30.2	41.5	47.3	56.0	62.6	68.8	67.1	60.8	51.6	40.1	29.6
	10%	DB	28.9	27.7	40.3	56.6	70.5	75.4	79.4	79.6	70.8	56.8	40.9	28.2
		MCWB	27.2	25.9	36.5	45.4	54.0	61.6	67.1	65.9	58.9	48.9	37.2	26.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.0	35.9	55.5	53.9	65.4	71.9	76.3	73.7	70.7	60.3	51.0	35.6
		MCDB	38.1	39.6	59.8	68.0	80.2	79.2	84.0	85.1	81.8	72.2	60.2	39.8
	2%	WB	33.3	33.2	45.8	50.7	62.0	67.9	72.9	71.3	66.2	56.2	44.2	32.3
		MCDB	35.7	35.3	53.2	63.1	71.3	75.9	81.8	82.1	74.9	65.4	49.6	34.7
	5%	WB	30.7	30.2	40.9	48.1	58.6	65.5	70.8	69.4	63.0	52.6	40.4	29.8
		MCDB	32.8	32.3	46.9	60.1	70.1	74.3	79.8	79.9	72.0	61.6	44.9	31.6
	10%	WB	27.1	25.9	36.6	45.6	56.1	63.4	68.6	67.1	60.5	49.1	37.3	26.9
		MCDB	28.8	27.8	40.3	55.5	67.0	72.2	77.2	76.9	69.0	56.5	40.7	28.3
Mean Daily Temperature Range	5% DB	MDBR	16.9	18.1	18.3	21.0	22.6	20.6	21.5	24.1	21.8	18.3	15.7	14.9
		MCDBR	17.9	18.5	23.1	30.2	31.2	25.2	25.2	29.7	29.7	27.3	22.2	17.2
		MCWBR	15.8	15.4	15.6	16.2	14.0	12.7	13.5	14.0	14.4	15.7	15.1	14.8
	5% WB	MCDBR	18.4	18.2	22.8	27.0	23.5	21.0	22.3	25.3	23.4	24.6	20.6	16.3
		MCWBR	16.4	15.5	15.6	15.4	12.1	12.5	13.6	13.2	13.4	14.4	14.5	14.1
Clear-Sky Solar Irradiance	taub	0.261	0.271	0.291	0.351	0.366	0.373	0.382	0.380	0.345	0.295	0.266	0.242	
	taud	2.282	2.271	2.384	2.334	2.350	2.369	2.351	2.349	2.432	2.545	2.480	2.391	
	Ebn at Noon	252	279	294	284	283	281	276	271	271	271	254	249	
	Edh at Noon	22	29	31	36	37	37	37	36	30	22	18	17	
All-Sky Solar Radiation	RadAvg	467	841	1244	1587	1759	1892	2007	1649	1201	733	411	335	
	RadStd	37	65	99	109	140	126	84	111	89	94	56	39	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (23 neighbors)	N/A	N/A	N/A	N/A	N/A	-1.22	N/A	N/A	N/A	N/A

Nomenclature: See separate page